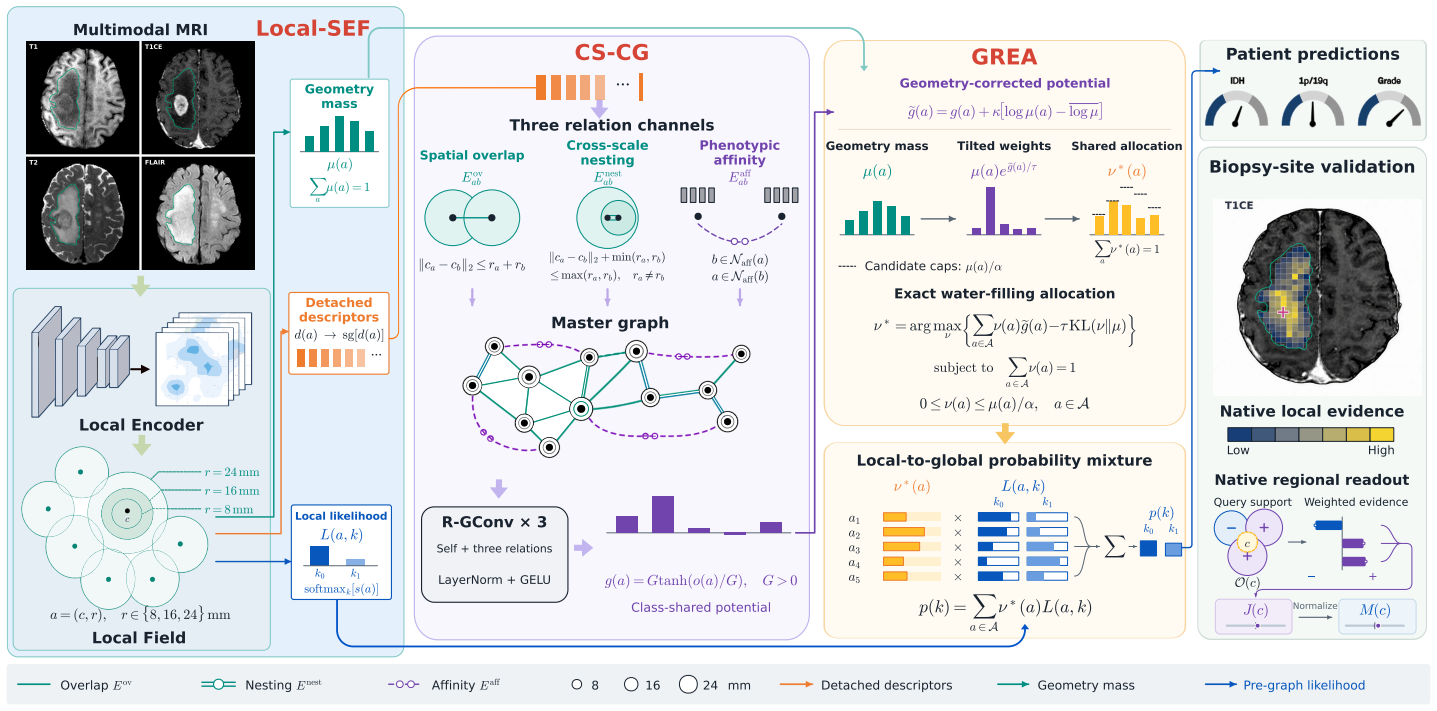

Multimodal MRI
T1
T1CE
T2
FLAIR
Local-SEF
Local Encoder
Local Field
$r = 24\,\mathrm{mm}$
$r = 16\,\mathrm{mm}$
$r = 8\,\mathrm{mm}$
$a = (c, r), \quad r \in \{8, 16, 24\}\,\mathrm{mm}$
Geometry mass
$\mu(a)$
$\sum_a \mu(a) = 1$
Detached descriptors
$d(a) \rightarrow \mathrm{sg}[d(a)]$
Local likelihood
$L(a,k)$
$\mathrm{softmax}_k[s(a)]$
CS-CG
Three relation channels
Spatial overlap
E^{ov}_{ab}
$\|c_a - c_b\|_2 \le r_a + r_b$
Cross-scale nesting
E^{nest}_{ab}
$\|c_a - c_b\|_2 + \min(r_a, r_b)$
$\le \max(r_a, r_b), \quad r_a \neq r_b$
Phenotypic affinity
E^{aff}_{ab}
$b \in \mathcal{N}_{\mathrm{aff}}(a)$
$a \in \mathcal{N}_{\mathrm{aff}}(b)$
Master graph
R-GConv × 3
Self + three relations
LayerNorm + GELU
$g(a) = G\tanh(o(a)/G), \quad G > 0$
Class-shared potential
GREA
Geometry-corrected potential
$\tilde{g}(a) = g(a) + \kappa[\log \mu(a) - \overline{\log \mu}]$
Geometry mass
Tilted weights
Shared allocation
$\mu(a)$
$\mu(a)e^{\tilde{g}(a)/\tau}$
$\nu^*(a)$
$\sum_a \nu^*(a) = 1$
Candidate caps: $\mu(a)/\alpha$
Exact water-filling allocation
$\nu^* = \arg\max_\nu \left\{ \sum_{a \in \mathcal{A}} \nu(a)\tilde{g}(a) - \tau \mathrm{KL}(\nu \| \mu) \right\}$
subject to $\sum_{a \in \mathcal{A}} \nu(a) = 1$
$0 \le \nu(a) \le \mu(a)/\alpha, \quad a \in \mathcal{A}$
Local-to-global probability mixture
$\nu^*(a)$
$L(a,k)$
$p(k)$
$p(k) = \sum_{a \in \mathcal{A}} \nu^*(a) L(a,k)$
Patient predictions
IDH
1p/19q
Grade
Biopsy-site validation
T1CE
Native local evidence
Low
High
Native regional readout
Query support
Weighted evidence
$\mathcal{O}(c)$
$J(c)$
Normalize
$M(c)$
Overlap E^{ov}
Nesting E^{nest}
Affinity E^{aff}
8
16
24 mm
Detached descriptors
Geometry mass
Pre-graph likelihood